Numerical Linear Algebra II: 06 Singular Value Decomposition

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Ostrava

Singular Value Decomposition

Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$, then $\boldsymbol{A}^T \boldsymbol{A}$ has eigenvalues and orthonormal eigenvectors:

$$\boldsymbol{A}^T \boldsymbol{A}\, \boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i, \quad ||\boldsymbol{v}_i|| = 1, \quad i = 1, 2, \ldots, n.$$

Without loss of generality, we assume the ordering of eigenvalues

$$\lambda_1 \geq \ldots \geq \lambda_r > 0, \quad \lambda_{r+1} = \ldots = \lambda_n = 0, \quad \text{where } r = \operatorname{rank}(\boldsymbol{A}^T \boldsymbol{A}).$$

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$$\boldsymbol{A}(\boldsymbol{A}^T\boldsymbol{A}\,\boldsymbol{v}_i) = \lambda_i \boldsymbol{A}\boldsymbol{v}_i \quad \Rightarrow \quad (\boldsymbol{A}\boldsymbol{A}^T)(\boldsymbol{A}\boldsymbol{v}_i) = \lambda_i(\boldsymbol{A}\boldsymbol{v}_i)$$

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Normalize:

$$\boldsymbol{u}_i = \frac{\boldsymbol{w}_i}{\|\boldsymbol{w}_i\|} = \frac{\boldsymbol{A}\boldsymbol{v}_i}{\sqrt{\lambda_i}}$$

giving orthonormal eigenvectors $\{\boldsymbol{u}_i\}$ of $\boldsymbol{A}\boldsymbol{A}^T$.

Singular Value Decomposition

Orthonormally complete the set $\{\boldsymbol{u}_i\}, i = 1, \ldots, r$ by vectors $\boldsymbol{u}_j, j = r+1, \ldots, m$ spanning $\mathcal{N}\left(\boldsymbol{A}\boldsymbol{A}^T\right)$.

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which gives the matrix form of the **singular value decomposition**

$$\boldsymbol{A}\,\boldsymbol{V} = \boldsymbol{U}\boldsymbol{\Sigma} \Rightarrow \boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\,\boldsymbol{V}^T$$

where

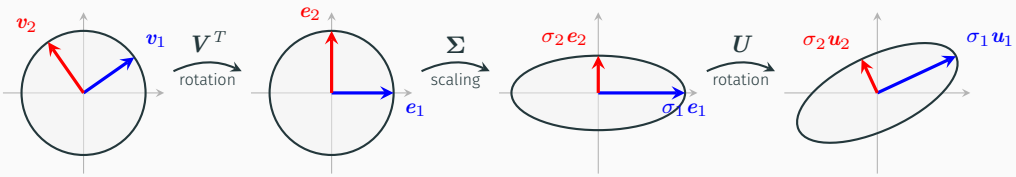
$$\boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_r & \boldsymbol{O} \\ \boldsymbol{O} & \boldsymbol{O} \end{pmatrix} \in \mathbb{R}^{m \times n}, \quad \boldsymbol{\Sigma}_r = \operatorname{diag}\left(\sigma_1, \ldots, \sigma_r\right),$$

and $\boldsymbol{U} = \left(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m\right)$, $\boldsymbol{V} = \left(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\right)$ are the **left** and **right singular vectors**.

Geometric Interpretation of SVD

Applying $\boldsymbol{Ax} = \boldsymbol{U\Sigma V}^T\boldsymbol{x}$ maps the unit sphere to an ellipsoid in three steps:

$$\boldsymbol{A} = \underbrace{\boldsymbol{U}}_{\text{rotation}} \cdot \underbrace{\boldsymbol{\Sigma}}_{\text{scaling}} \cdot \underbrace{\boldsymbol{V}^T}_{\text{rotation}}$$



Properties of SVD

Let $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T \in \mathbb{R}^{m \times n}$ with singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$.

- $\mathrm{rank}(\boldsymbol{A}) = r$, i.e., the number of non-zero singular values.

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- $\mathrm{rank}(\boldsymbol{A}) = r$, i.e., the number of non-zero singular values.
- The non-zero singular values σ_i are the square roots of the non-zero eigenvalues of $\boldsymbol{A}^T\boldsymbol{A}$ and $\boldsymbol{A}\boldsymbol{A}^T$:

$$\sigma_i = \sqrt{\lambda_i(\boldsymbol{A}^T\boldsymbol{A})} = \sqrt{\lambda_i(\boldsymbol{A}\boldsymbol{A}^T)}, \quad i = 1, \ldots, r.$$

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- If $\boldsymbol{A}$ is symmetric ($\boldsymbol{A} = \boldsymbol{A}^T$), then the singular values are the absolute values of the eigenvalues of $\boldsymbol{A}$:

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- The spectral norm (2-norm) of $\boldsymbol{A}$ is the largest singular value:

$$\|\boldsymbol{A}\|_2 = \sigma_1.$$

SVD and the Four Fundamental Subspaces

Let $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T \in \mathbb{R}^{m\times n}$ with $\operatorname{rank}(\boldsymbol{A}) = r$. The singular vectors provide **orthonormal bases** for all four fundamental subspaces of $\boldsymbol{A}$:

- **Column space (range)** of $\boldsymbol{A}$ ($\dim = r$):

$$\mathcal{R}(\boldsymbol{A}) = \operatorname{span}\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r\} \subseteq \mathbb{R}^m$$

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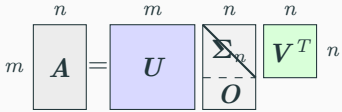
- **Null space (kernel)** of $\boldsymbol{A}$ ($\dim = n - r$):

$$\mathcal{N}(\boldsymbol{A}) = \operatorname{span}\{\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n\} \subseteq \mathbb{R}^n$$

Reduced Forms of SVD ($m \geq n \geq r$)

Full SVD

$$\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$$

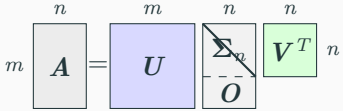


$\boldsymbol{U} \in \mathbb{R}^{m \times m}$, $\boldsymbol{\Sigma} \in \mathbb{R}^{m \times n}$, $\boldsymbol{V} \in \mathbb{R}^{n \times n}$

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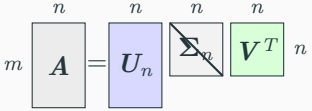
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Thin (Economy) SVD

$$\boldsymbol{A} = \boldsymbol{U}_n\boldsymbol{\Sigma}_n\boldsymbol{V}^T$$



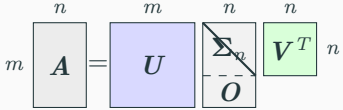
$\boldsymbol{U}_n \in \mathbb{R}^{m \times n}$, $\boldsymbol{\Sigma}_n \in \mathbb{R}^{n \times n}$, $\boldsymbol{V} \in \mathbb{R}^{n \times n}$

(drops zero rows of $\boldsymbol{\Sigma}$)

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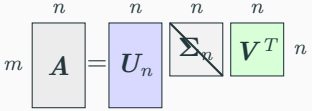
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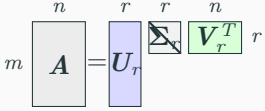


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(drops zero rows of $\boldsymbol{\Sigma}$)

Compact SVD

$$\boldsymbol{A} = \boldsymbol{U}_r\boldsymbol{\Sigma}_r\boldsymbol{V}_r^T$$



$\boldsymbol{U}_r \in \mathbb{R}^{m \times r}$, $\boldsymbol{\Sigma}_r \in \mathbb{R}^{r \times r}$, $\boldsymbol{V}_r \in \mathbb{R}^{n \times r}$

(keeps only $\sigma_i > 0$, $r = \mathrm{rank}(\boldsymbol{A})$)

SVD: Matrix A as a Sum of Rank-1 Matrices

Writing out the matrices explicitly:

$$\boldsymbol{A} = \underbrace{\begin{bmatrix} | & & | \\ \boldsymbol{u}_1 & \cdots & \boldsymbol{u}_r \\ | & & | \end{bmatrix}}_{\boldsymbol{U} \text{ (first } r \text{ cols)}} \underbrace{\begin{bmatrix} \sigma_1 & & 0 \\ & \ddots & \\ 0 & & \sigma_r \end{bmatrix}}_{\boldsymbol{\Sigma} \text{ (top-left } r \times r)} \underbrace{\begin{bmatrix} - & \boldsymbol{v}_1^T & - \\ & \vdots & \\ - & \boldsymbol{v}_r^T & - \end{bmatrix}}_{\boldsymbol{V}^T \text{ (first } r \text{ rows)}}$$

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Rank-1 Decomposition

Multiplying the columns of $\boldsymbol{U}$ by the rows of $\boldsymbol{V}^T$ scaled by σ:

$$\boxed{\boldsymbol{A} = \sum_{i=1}^{r} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T}$$

Each term $\boldsymbol{u}_i \boldsymbol{v}_i^T$ is an $m \times n$ matrix of rank 1 (outer product).

Eckart–Young–Mirsky Theorem

Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ and let $k \leq \text{rank}(\boldsymbol{A})$. Then

$$\boldsymbol{A}_k = \sum_{i=1}^{k} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T$$

is the best approximation of matrix $\boldsymbol{A}$ with $\text{rank}(\boldsymbol{A}_k) = k$, i.e.,

$$||\boldsymbol{A} - \boldsymbol{A}_k||_2 = \min_{\substack{\widehat{\boldsymbol{A}} \in \mathbb{R}^{m \times n} \\ \text{rank}(\widehat{\boldsymbol{A}}) \leq k}} ||\boldsymbol{A} - \widehat{\boldsymbol{A}}||_2 = \sigma_{k+1}$$

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Note that

$$||\boldsymbol{A} - \boldsymbol{A}_k||_2 = \left\| \sum_{i=1}^{r} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T - \sum_{i=1}^{k} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T \right\|_2 = \left\| \sum_{i=k+1}^{r} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T \right\|_2 = \sigma_{k+1}$$

Moore–Penrose Pseudoinverse

For $\boldsymbol{A} \in \mathbb{R}^{m \times n}$, the **Moore–Penrose pseudoinverse** $\boldsymbol{A}^{+} \in \mathbb{R}^{n \times m}$ is the unique matrix satisfying the four Penrose conditions:

$$\boldsymbol{A}\boldsymbol{A}^{+}\boldsymbol{A} = \boldsymbol{A}, \quad \boldsymbol{A}^{+}\boldsymbol{A}\boldsymbol{A}^{+} = \boldsymbol{A}^{+}, \quad (\boldsymbol{A}\boldsymbol{A}^{+})^{T} = \boldsymbol{A}\boldsymbol{A}^{+}, \quad (\boldsymbol{A}^{+}\boldsymbol{A})^{T} = \boldsymbol{A}^{+}\boldsymbol{A}.$$

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Computation via SVD: Given $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^{T}$ with $\operatorname{rank}(\boldsymbol{A}) = r$,

$$\boxed{\boldsymbol{A}^{+} = \boldsymbol{V}\boldsymbol{\Sigma}^{+}\boldsymbol{U}^{T} = \boldsymbol{V}_{r}\boldsymbol{\Sigma}_{r}^{-1}\boldsymbol{U}_{r}^{T} = \sum_{i=1}^{r} \frac{1}{\sigma_i}\boldsymbol{v}_i\boldsymbol{u}_i^{T}}$$

where $\boldsymbol{\Sigma}^{+} \in \mathbb{R}^{n \times m}$ is formed by transposing $\boldsymbol{\Sigma}$ and taking the reciprocal of non-zero singular values:

$$\boldsymbol{\Sigma}^{+} = \begin{pmatrix} \boldsymbol{\Sigma}_{r}^{-1} & \boldsymbol{O} \\ \boldsymbol{O} & \boldsymbol{O} \end{pmatrix} \in \mathbb{R}^{n \times m}, \quad \boldsymbol{\Sigma}_{r}^{-1} = \operatorname{diag}\left(\frac{1}{\sigma_1}, \ldots, \frac{1}{\sigma_r}\right).$$

Least-Squares Problems via Moore–Penrose Pseudoinverse

Given $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ and $\boldsymbol{b} \in \mathbb{R}^{m}$, consider the linear least-squares problem:

$$\min_{\boldsymbol{x} \in \mathbb{R}^{n}} \|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|_2.$$

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Using the SVD $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ ($\operatorname{rank}(\boldsymbol{A}) = r$) and substituting $\boldsymbol{y} = \boldsymbol{V}^T\boldsymbol{x} \iff \boldsymbol{x} = \boldsymbol{V}\boldsymbol{y} = \sum_{i=1}^n y_i \boldsymbol{v}_i$:

$$\begin{aligned} \|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|_2^2 &= \|\boldsymbol{U}^T(\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T\boldsymbol{x} - \boldsymbol{b})\|_2^2 = \|\boldsymbol{\Sigma}\boldsymbol{y} - \boldsymbol{U}^T\boldsymbol{b}\|_2^2 \\ &= \sum_{i=1}^{r} \left(\sigma_i y_i - \boldsymbol{u}_i^T\boldsymbol{b}\right)^2 + \sum_{i=r+1}^{m} (\boldsymbol{u}_i^T\boldsymbol{b})^2. \end{aligned}$$

The minimum is achieved when

$$y_i = \frac{\boldsymbol{u}_i^T\boldsymbol{b}}{\sigma_i} \quad \text{for } i = 1, \ldots, r,$$

while $y_{r+1}, \ldots, y_n$ do not affect the residual and can be chosen arbitrarily.

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while $y_{r+1}, \ldots, y_n$ do not affect the residual and can be chosen arbitrarily.

Least-Squares Solutions via Moore–Penrose Pseudoinverse

Writing the free coefficients as $y_i = \boldsymbol{v}_i^T \boldsymbol{w}$ $(i > r)$ for an arbitrary vector $\boldsymbol{w} \in \mathbb{R}^n$, and using $\boldsymbol{I} = \boldsymbol{V}\boldsymbol{V}^T = \sum_{i=1}^{n} \boldsymbol{v}_i \boldsymbol{v}_i^T$ and $\boldsymbol{A}^+\boldsymbol{A} = \sum_{i=1}^{r} \boldsymbol{v}_i \boldsymbol{v}_i^T$, all least-squares solutions are given by:

$$\begin{aligned}\boldsymbol{x} &= \sum_{i=1}^{r} \frac{\boldsymbol{u}_i^T \boldsymbol{b}}{\sigma_i} \boldsymbol{v}_i + \sum_{i=r+1}^{n} (\boldsymbol{v}_i^T \boldsymbol{w}) \boldsymbol{v}_i \\ &= \left(\sum_{i=1}^{r} \frac{1}{\sigma_i} \boldsymbol{v}_i \boldsymbol{u}_i^T \right) \boldsymbol{b} + \left(\sum_{i=r+1}^{n} \boldsymbol{v}_i \boldsymbol{v}_i^T \right) \boldsymbol{w} = \boldsymbol{A}^+ \boldsymbol{b} + (\boldsymbol{I} - \boldsymbol{A}^+ \boldsymbol{A}) \boldsymbol{w}.\end{aligned}$$

Since $\|\boldsymbol{x}\|_2^2 = \|\boldsymbol{y}\|_2^2 = \sum_{i=1}^{r} y_i^2 + \sum_{i=r+1}^{n} y_i^2$, setting $y_i = 0$ for $i > r$ $(\boldsymbol{w} = \boldsymbol{0})$ yields the unique **minimum-norm least-squares solution** $(\min \|\boldsymbol{x}\|_2)$:

$$\boxed{\boldsymbol{x}_{\mathrm{LS}} = \boldsymbol{A}^+ \boldsymbol{b} = \boldsymbol{V} \boldsymbol{\Sigma}^+ \boldsymbol{U}^T \boldsymbol{b} = \sum_{i=1}^{r} \frac{\boldsymbol{u}_i^T \boldsymbol{b}}{\sigma_i} \boldsymbol{v}_i}$$

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